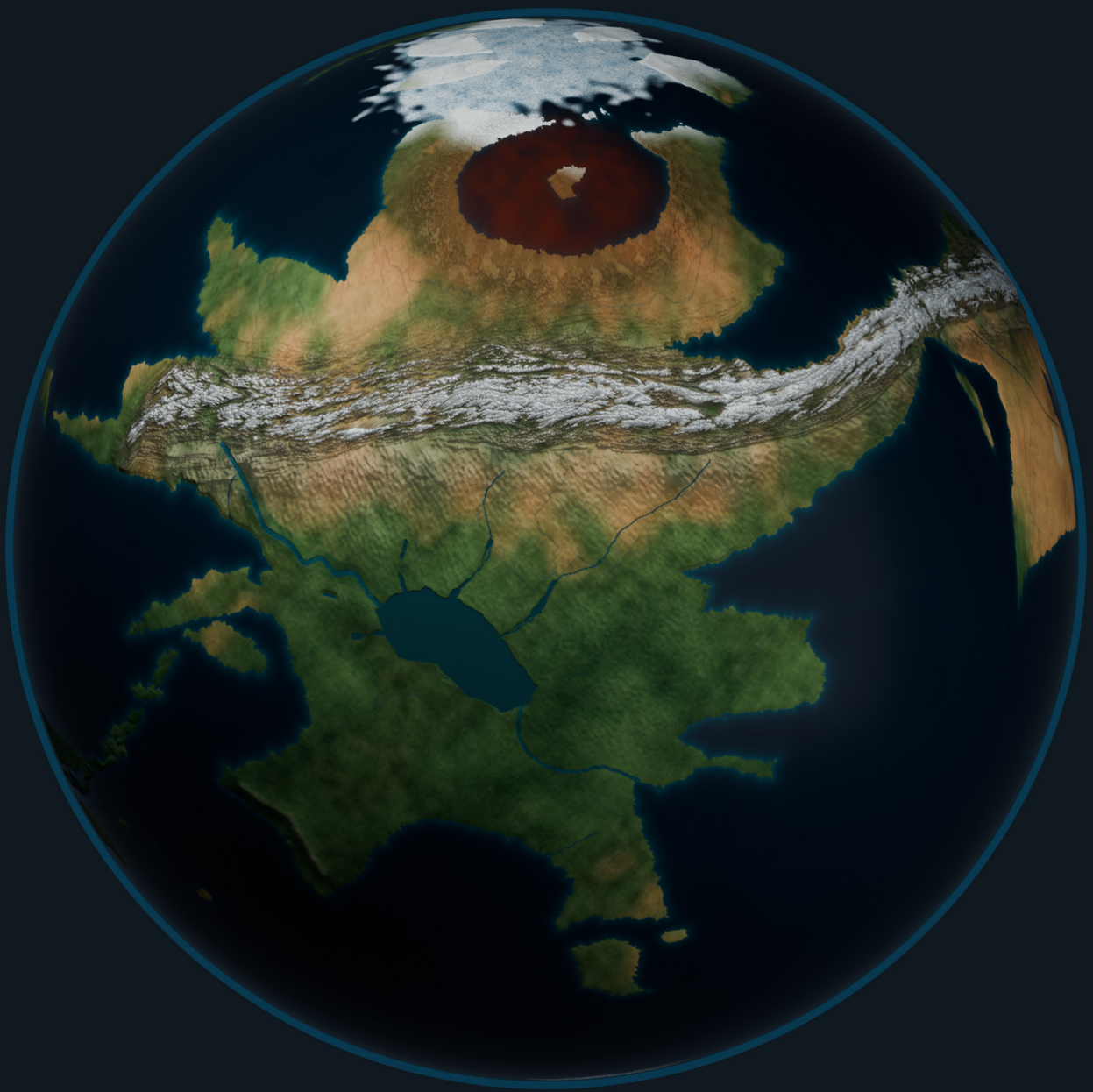
CATWORLD
THE WORLD MODEL

Creator & showrunner field guide



A planet to draw from. A geography to tell stories in.

Revision 4 / 31 October 2026

Three connected continental cores, a northern impact province,
the great reservoir, long seasons and a rebuilt far-side continent.

READING ROUTE

A guide to the model

Read the geography first, then use the reference plates. Established creator facts, draft-deck material and new cartographic proposals are distinguished throughout. Square-bracket source keys point to the reference notes.

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What is established, and what is proposed

Confirmed by the creator brief

Terra is approximately Earth-sized, with Earth-like axial tilt and a 4,000-Earth-year orbital period. Three major continental cores connect through gargantuan mountain belts. A huge northern impact brought metals to an iron-poor world. The red basin, continental reservoir and exceptional hydroelectric dam are central features.

Confirmed layout corrections

The main Panterran outline remains recognizable. The mountain bridge is shorter. The green far-side core moves northward and loses its long southern tail. In the marked far-side view, the large left interior is dry and the right maritime country is wet. A long island chain can represent the old connection toward Catworld country.

Confirmed water order

Seattle is upstream: its major river flows INTO the super-lake near London and Rome. The dam controls the lake outlet. The large city lies downstream. This guide uses Amsterdam for that city, following the pitch deck; city names are working shorthand.

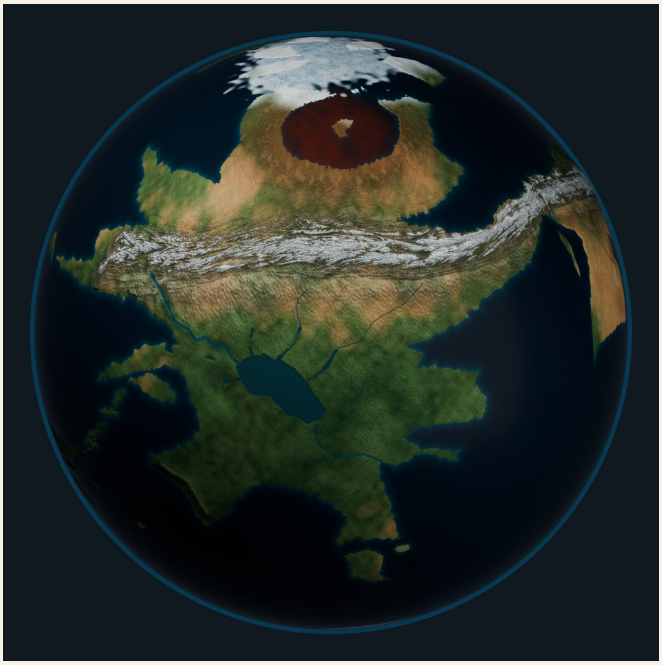
Cartographic proposals

The exact 6,100 km radius, 23.4-degree tilt, 24-hour day, present northern-autumn epoch, coordinates, tectonic history, wind routes, ice margins and most descriptive place names are design proposals. The model is a coherent interpretation of the sketches, not a uniquely recoverable projection.

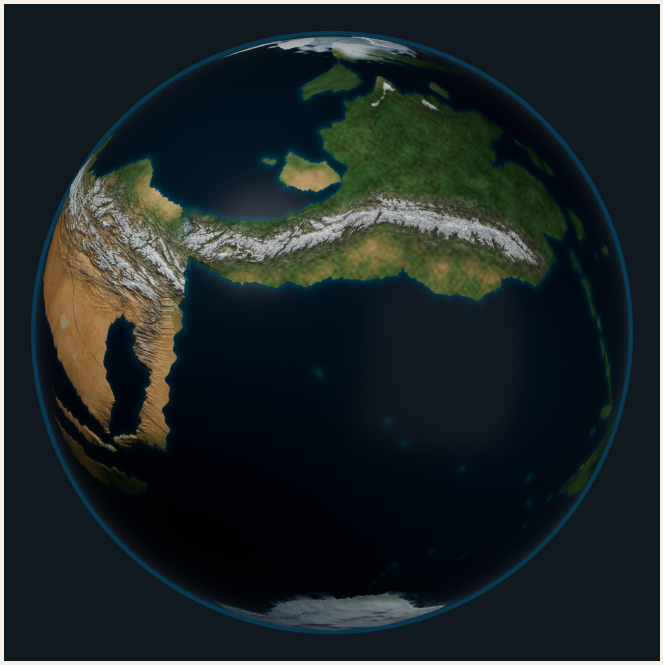
Open worldbuilding decisions

The stellar/heating explanation for the long orbit, exact chronology, final political borders, city names, reservoir history and reliable power output remain open. The deck is a development draft. Its notes supply context; they do not override the creator's latest corrections.

The planet at a glance



PANTERRA / FRONT



FAR-SIDE HEMISPHERE

One connected inhabited system

Panterra anchors the principal drawing hemisphere. A shorter mountain neck reaches the eastern dry core. A second oblique junction connects that core to the maritime continent on the far side. These are three old continental interiors within one connected land system, not three unrelated islands.

A different far-side silhouette

The maritime continent now occupies the northern ocean area marked in the redline. Its broad interior lies north of the main collision range. A short, irregular southern foreland replaces the long green tail. Western bays, an eastern marginal gulf and a cool northern peninsula give it distinct edges.

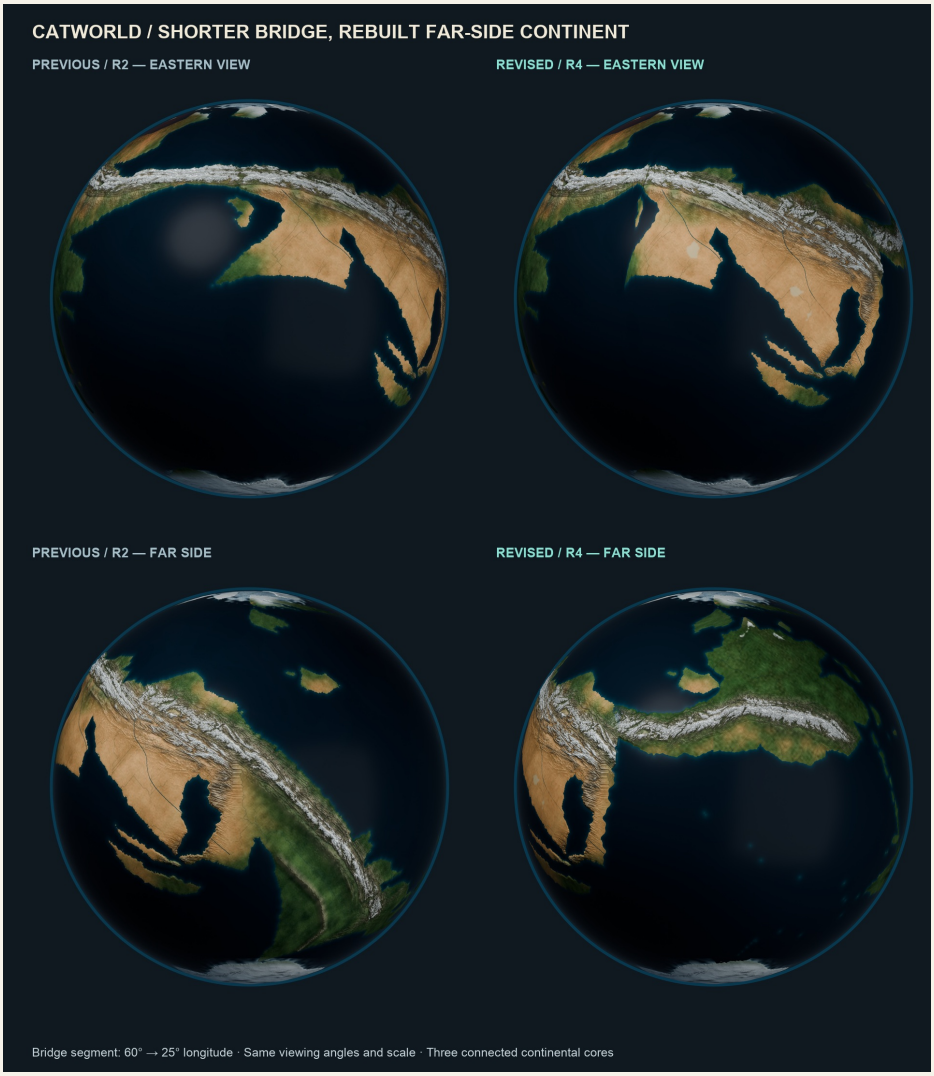
Water and the poles

Large oceans separate the inhabited continental system from a separate south-polar landmass. The north is chiefly ocean with islands and broken pack ice. The island bridge curves through the western ocean toward Panterra. Its gaps matter: they create maritime routes and possible former shelf crossings.

Orientation rule

Longitude is an arbitrary model coordinate. Front = 0 degrees, east = +90, west = -90, far side = 180. The two images above are opposite sides of the same globe; their shorelines and terrain agree.

What changed in this pass



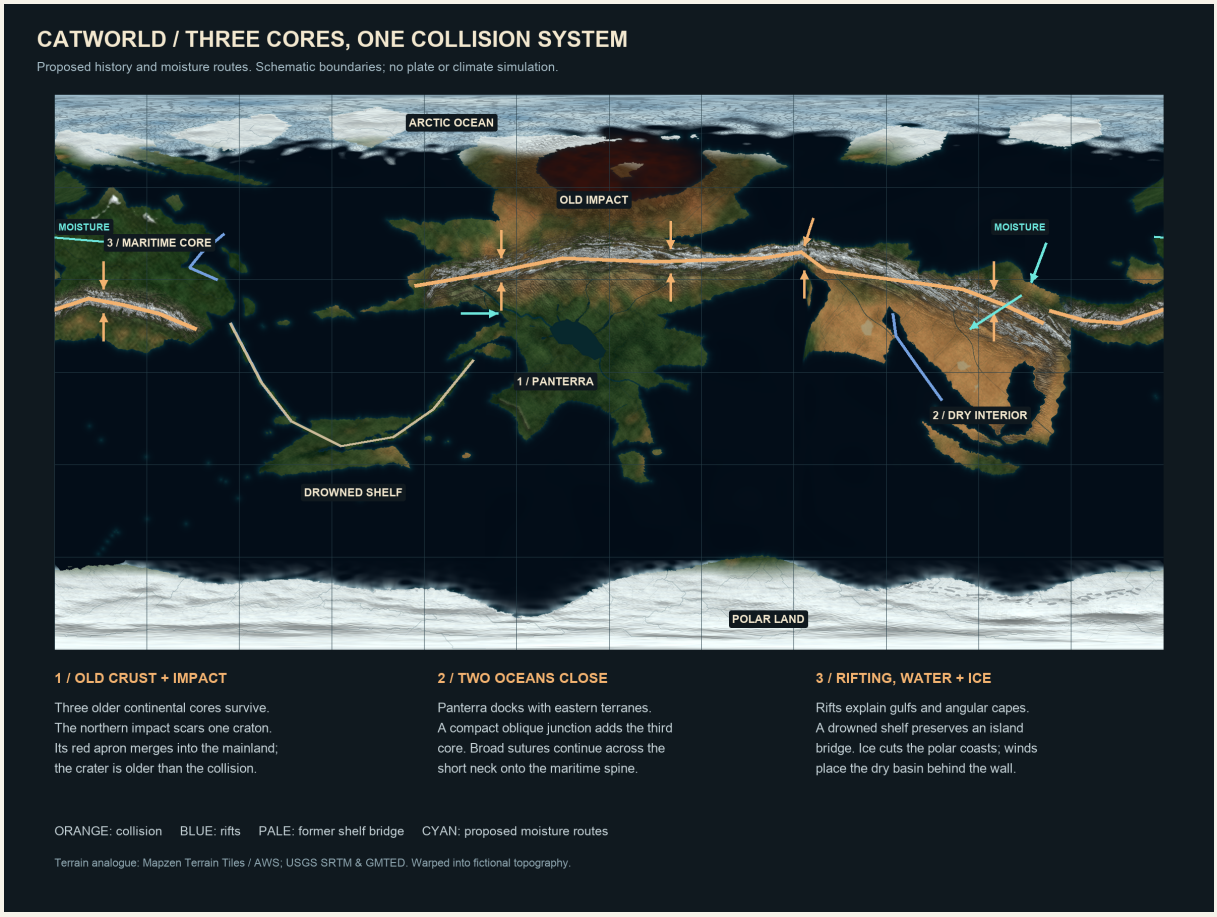
Shorter connection, fuller continent

The marked eastern bridge segment shrinks from 60 to 25 degrees of longitude, about 58%. This is a measurement of the chosen bridge endpoints, not the entire mountain system. Panterra stays in place while the adjacent core moves closer.

The far side is rebuilt

The maritime core now has a substantial northern interior, an old upland shield, a folded southern range and branching drainage. It is no longer a rotated strip with a long tail. The desert has differentiated dune fields and pale salt basins. The island chain, both polar views and the Seattle waterway remain part of the review.

Why these continents fit together



Stage 1: older crust and a northern impact

Three older continental blocks exist before the present mountain system. The northern impact damages one of them, leaving a basin, fractured rock and a broad ejecta/mineral province. The crater and the later long mountain wall have different origins.

Stage 2: successive collisions

Oceanic crust is consumed between Panterra and eastern terranes. Continental collision thickens and folds the crust. A later oblique collision docks the maritime block against the eastern core. The two sutures join, producing a connected, branching mountain system rather than independent decorative ridges. [S1]

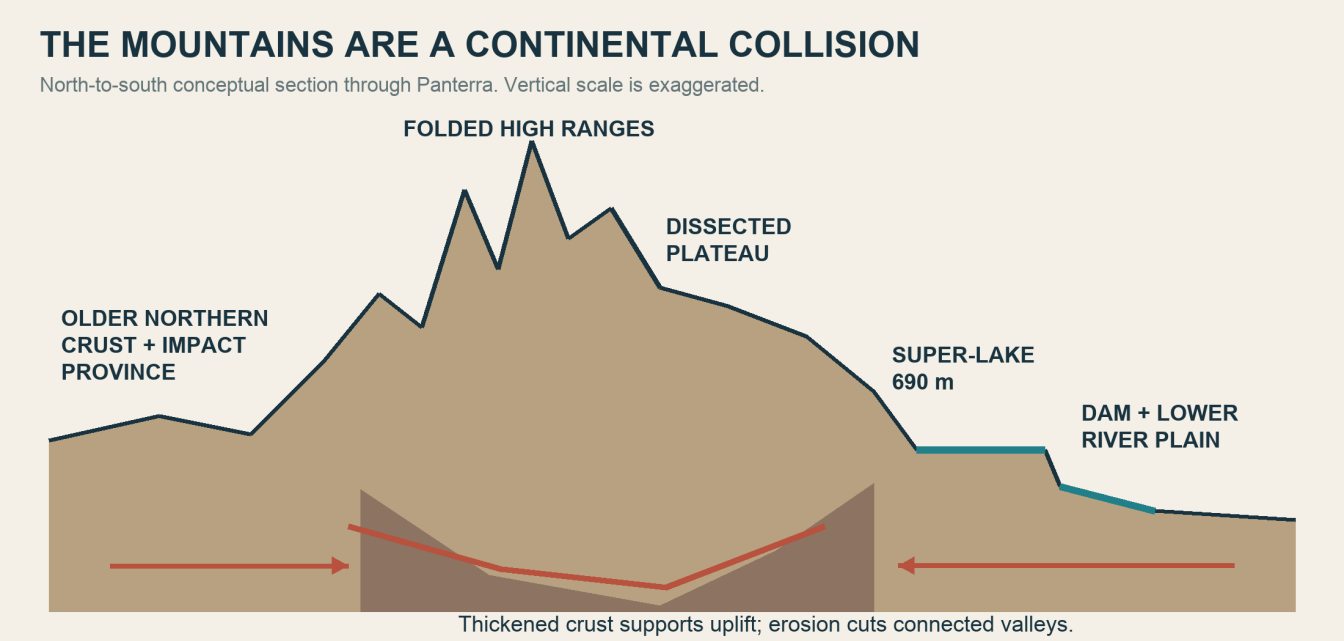
Stage 3: erosion and partial reopening

Rivers cut the high ground. Old faults and later extension guide the large gulfs, capes and island shelf. Short mountain necks remain where uplifted crust bridges the older continental interiors. Coastline angles follow these broad structural directions.

What the diagram means

The lines are proposed sutures and rifts, not a solved global plate network. No plate velocities, absolute ages or mantle dynamics have been calculated. A future reconstruction can add ocean ridges and transform faults without changing the central story geography.

Gargantuan ranges, not a row of peaks



A broad elevated system

The high country includes parallel folded ridges, interior plateaus, dissected foothills and deep transverse valleys. Proposed peaks approach 10.8 km. The high belt is several hundred kilometres wide, with a broader surrounding mountain province. Those dimensions are deliberate fantasy-world choices.

Why the plateau sits below it

Thickened crust raises a wide region. Erosion then cuts the lower flank into valleys and terraces, creating the transition from the Northern Divide to the Panterran Plateau and reservoir basin. The lake occupies a depression within that landscape, rather than resting on an arbitrary flat shelf.

Why the terrain looks structured

A real Himalayan elevation analogue supplies ridge-and-valley texture, warped into the invented belt. The far-side range is rebuilt along its own suture and blended into the compact junction. The source elevations are modified reference data, not a claim that Terra copies Earth.

Reading the height correctly

The physical elevation field is in metres. The globe displays relief at TEN times its radial height so mountains remain readable from space. The cross-section is also schematic. Neither image should be used to infer literal cliff heights or structural dimensions.

Panterra: the principal story landscape



A connected sequence of places

The Northern Divide forms the high northern boundary. Below it lie a dissected plateau, wet western uplands, the Seattle river corridor, the great lake and cultivated southern lowlands. The downstream metropolis occupies the outlet trade system. These relationships matter more than any provisional name.

Forests, fields and industry

Wet upland valleys support giant old-growth forests and timber settlements. Lower gradients and water access support farming near the lake and river plains. Logging, milling, canals and electricity connect those resource zones to cities. The deck's giant-tree and industrial themes therefore occupy specific parts of the terrain. [D4, D41, D47]

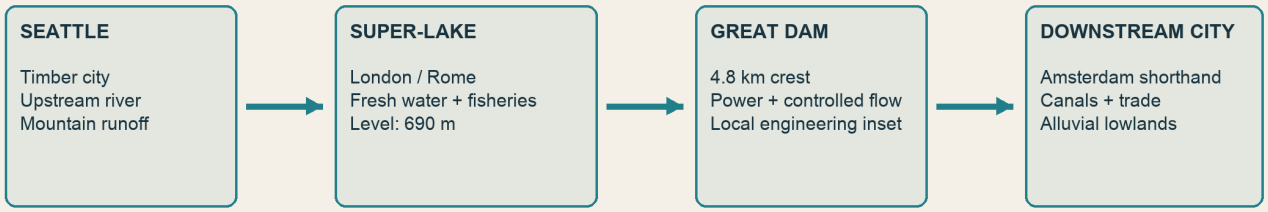
Defence and access

The mountains and indented shores limit easy overland approaches. Gate valleys offer supply corridors through the range. Ports and river mouths provide alternatives, so the empire is difficult to enter with heavy freight without becoming implausibly sealed off from the world.

Seattle flows into the lake

THE WATERWAY HAS ONE CLEAR DIRECTION

Seattle is upstream of the lake. The large trading city is downstream of the dam.



Water, logs and cargo follow the corridor. Rapids, locks, canals and transshipment need local design before the whole route can be treated as navigable.

Upstream means upstream

Mountain tributaries gather above Seattle. The primary channel then descends through the western foreland into the reservoir. Its sampled water surface decreases toward 690 m. The lake does not drain west to Seattle. The outlet instead leaves through the dam toward the downstream trading city.

The river is deliberately visible

The main waterway is wider in the texture than the secondary drainage so it remains useful for drawing. At a 4K global texture, a few pixels represent many kilometres: this is a reference line-width exaggeration, not a literal river width. The main channel has a separately graded surface and a recorded downhill check.

Transport requires local decisions

Logs and cargo can follow this corridor, but an upstream mountain river is not automatically navigable end to end. Rapids, bypass canals, locks, portage or transshipment belong at abrupt gradients. Establish those local links before staging an uninterrupted boat journey.

Other drainage

The maritime core has a new elevation-routed tributary network, with grid corners smoothed for readability. Secondary rivers and desert wadis are schematic at this scale. The check on the primary Seattle route does not certify every tributary or predict floods.

A continental lake needs a continental basin

What the dam can actually do

The model reservoir covers about 1.54 million square kilometres. A dam can control a lake this large only if a naturally enclosed basin already exists, with one dominant bedrock outlet. The dam raises and stabilizes water in that basin; it does not create the depression itself.

How a small lake becomes a super-lake

A small predecessor lake can occupy the deepest part of a broad valley system. Raising its outlet floods extensive shallow land and connected tributary valleys. The model proposes that history, but does not yet contain a surveyed pre-dam terrain surface. Shoreline benches and drowned valleys are the visual evidence to carry into local maps.

Water budget, explicitly assumed

For an illustrative 6 million km2 catchment, 0.4 m of annual runoff supplies 2,400 km3/year. Net evaporation of 0.8 m/year across the super-lake removes roughly 1,235 km3/year. That leaves about 1,165 km3/year before other losses and uses. These are scenario inputs, not measured or climate-simulated facts.

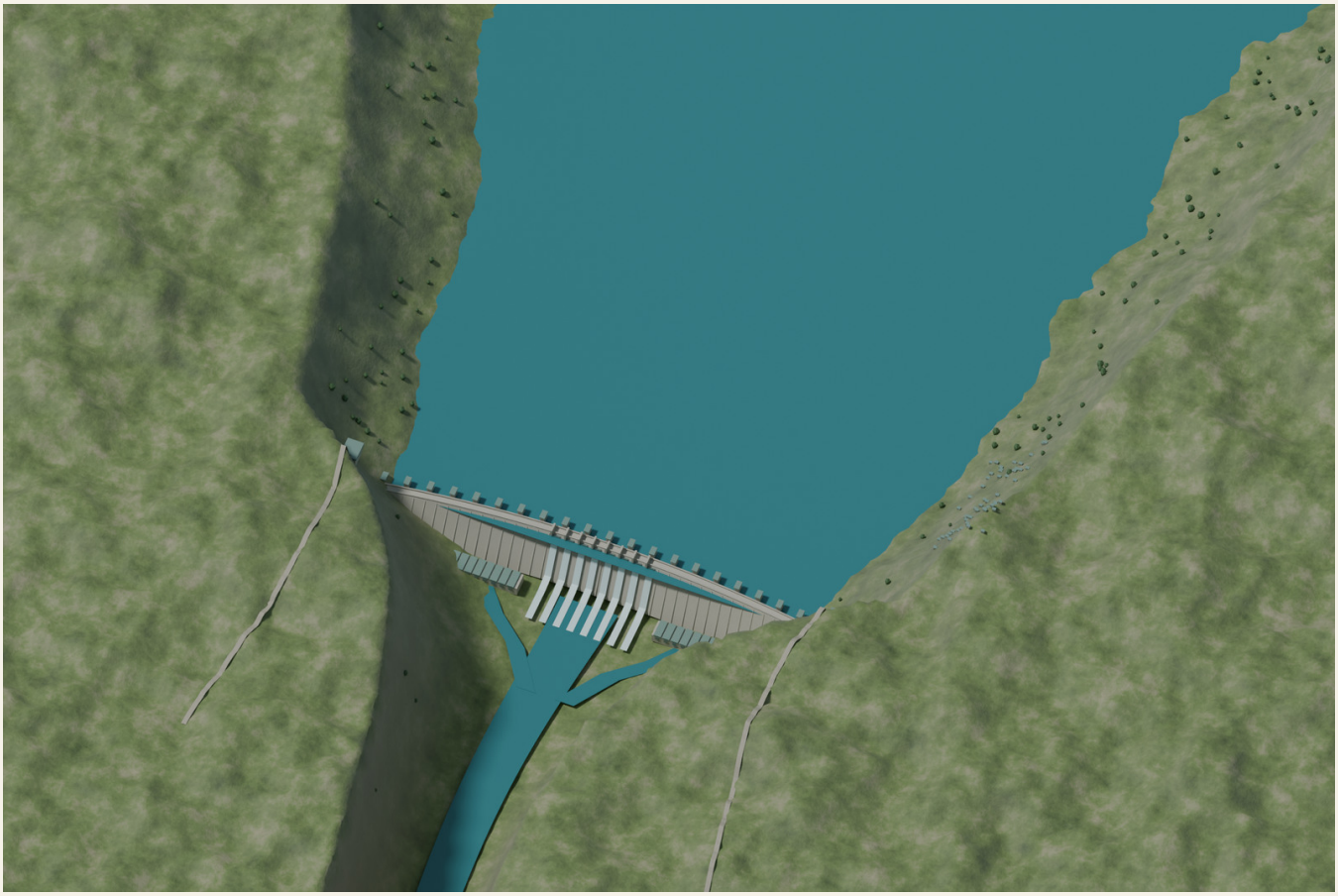
The filling history matters

At an assumed 100 m mean depth, storage would be about 154,000 km3. Even a net filling flow of 25,000 m3/s would take roughly two centuries from empty. A credible history should allow existing water, staged impoundment and long filling periods, rather than an overnight transformation.

Why it remains fresh

A sustained controlled outlet exports dissolved salts. Large inflow and continued flushing support fresh water; a closed, evaporation-dominated basin would tend toward salinity. Sediment trapping, shoreline flooding and changes in downstream fertility are important consequences for settlements and politics.

The Great Panterran Dam



The designed object

The separate inset has a 4.8 km crest, approximately 470 m structural height, reservoir level at 690 m and crest near 715 m. It includes intake towers, bronze gates, spillways, turbine halls, tailraces, access roads, checkpoints and a fishing village. One inset unit equals one kilometre.

Power comes from head and flow

Using P = density x gravity x flow x head x efficiency, an assumed 10,000 m3/s, 350 m net head and 90% efficiency gives about 31 GW. A nominal 60 GW would require roughly 19,400 m3/s at that head. Lake size alone cannot guarantee that discharge. [S7]

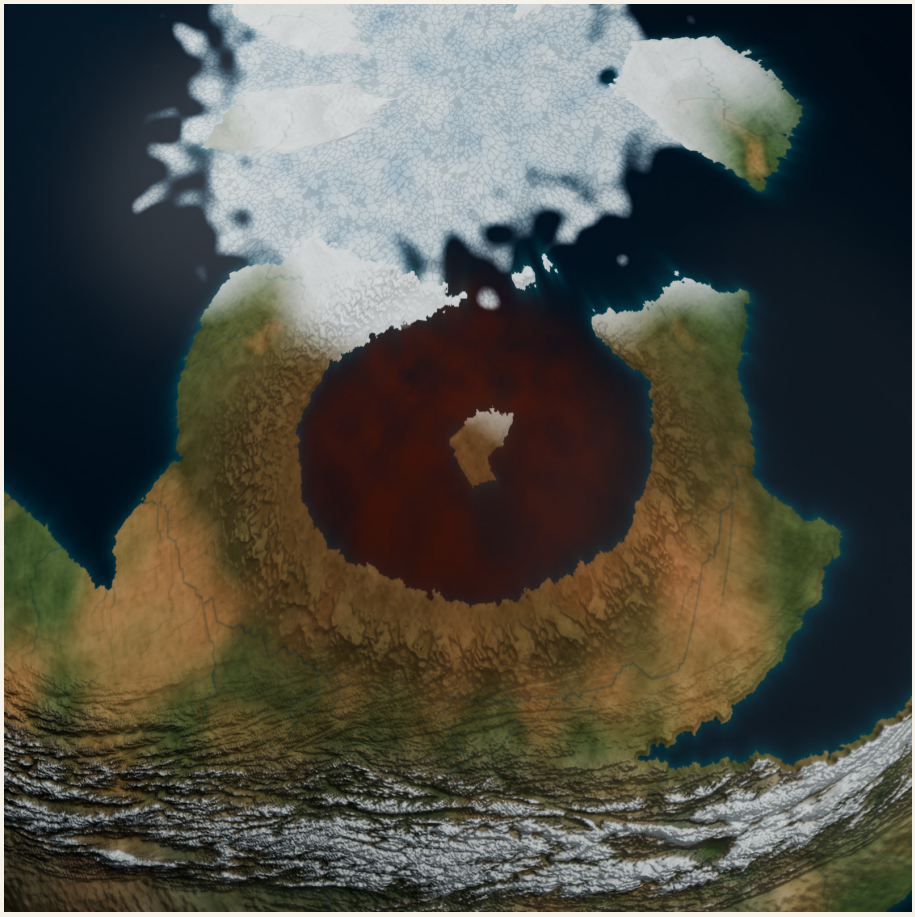
Scale and limits

The concept is larger in structural height than Hoover (about 221 m) and Three Gorges (181 m). Its illustrative 31 GW exceeds Three Gorges' 22.5 GW installed capacity. These comparisons establish ambition, not engineering feasibility. The inset is a fictional design study, not a construction-ready dam. [S8, S9]

Global and local scales

The dam wall is below the globe texture resolution. The local study shows the abrupt drop; the global river grade is smoothed. The village terrace is resolved in the inset, not in its approximate global anchor. Do not measure the local gorge from the planetary mesh.

The northern impact and the red sea



An old scar incorporated into a continent

The basin has a central uplift, a broken rim and a broad apron that merges into the northern mainland. It is older than the collision belt. Erosion and subsequent tectonics modify its edges, so it reads as part of the crust rather than a circular object placed on top.

Why the red works

Rust-coloured soil and sediment suggest oxidized iron minerals such as hematite. Red water represents suspended mineral-rich material and staining, not liquid metal. Colour is a strong visual cue, but it does not prove that the iron came from a meteorite. [S3]

Ore and the story

The iron-rich northern province is a creator premise. Impact melting, fractured rock and later fluid circulation offer a proposed way to redistribute and concentrate resources. Real impact structures can host ore deposits, but rare earths and enriched uranium are not automatic products of an iron-rich impact. Keep those deck claims as separate fiction decisions. [S4; D5, D47]

Time to recover

A country-scale impact implies an ancient catastrophe. Its age must allow climate and ecosystems to recover before the current bird kingdoms. The exact age, original diameter and ore inventory remain unspecified.

The dry eastern core



The corrected dry side

This is the broad dry interior on the left of the creator's far-side view. High northern and eastern mountain barriers intercept proposed moisture-bearing air. Downwind descent and distance from the ocean reinforce aridity. A narrow wetter coastal margin can coexist with a huge inland desert.

More than one sand colour

The new pass distinguishes dune fields, gravel country, pale salt basins, mountain-fed channels and rocky rift margins. Permanent rivers can cross dry land when fed from snowy highlands, while smaller wadis flow episodically. The salt pans are proposed landscape details, not named canonical locations.

Why the coast has edges

Deep gulfs follow extensional weaknesses and old faults. Long peninsulas and offshore remnants preserve the direction of the older crust. The shortened western connection brings this core closer to Panterra without reducing the mountain system to a narrow footbridge.

Story use

The reference material associates eastern deserts with reptile polities. Terrain can guide caravan corridors, mountain crossings, water-dependent settlements and trading ports. Their borders and cultures are not fixed by biome alone. [D4, D42]

The maritime continent, rebuilt



A real interior above the range

The former long green strip is replaced by a compact northern-reaching continent. Its main collision belt lies along the southern margin. North of it are forested river country and an older upland shield; farther north, a peninsula grades toward cool woodland and tundra.

Coasts with different histories

The western embayments face an open marginal sea. An eastern gulf cuts back into the mainland between angular headlands. Offshore fragments and the beginning of the island chain continue those structural directions. These features give the coast readable regional identities.

A coherent relief hierarchy

The young southern range is the highest terrain. The old shield is lower and more weathered. Smaller northern uplands catch snow at lower elevations in the colder latitude band. Rivers gather in the intervening lowlands and reach the nearby seas.

A role without invented canon

This core can supply forests, fisheries and ports beyond Panterra, with its own routes and histories. The model does not automatically assign it to a faction or claim it is Old Panterra. That historical identity should be chosen by the creator after reviewing the new layout.

The island bridge and its missing land



A long broken connection

The chain curves from the maritime margin toward Panterra's western and southwestern approaches. Elongated islands, straits and shelf fragments follow a consistent trend. They replace a random scattering of identical round islands.

What a former land bridge requires

A past crossing needs shallow saddles between islands and a sufficiently lower sea level or different tectonic elevation. The surface chain is modeled; a complete revised bathymetric survey is not. Do not yet claim that every channel was dry at once or attach an exact migration date.

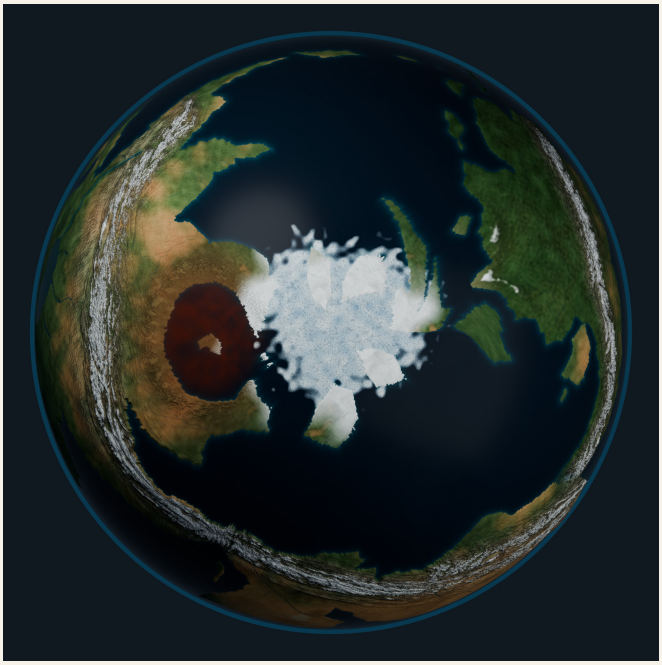
Ports and ruins belong at useful places

Sheltered sounds support fishing towns. Navigable straits create customs and resupply stations. Abandoned headlands can preserve road approaches or ruins above drowned valleys. Settlement density should follow usable harbours and freshwater access, not even spacing along the chain.

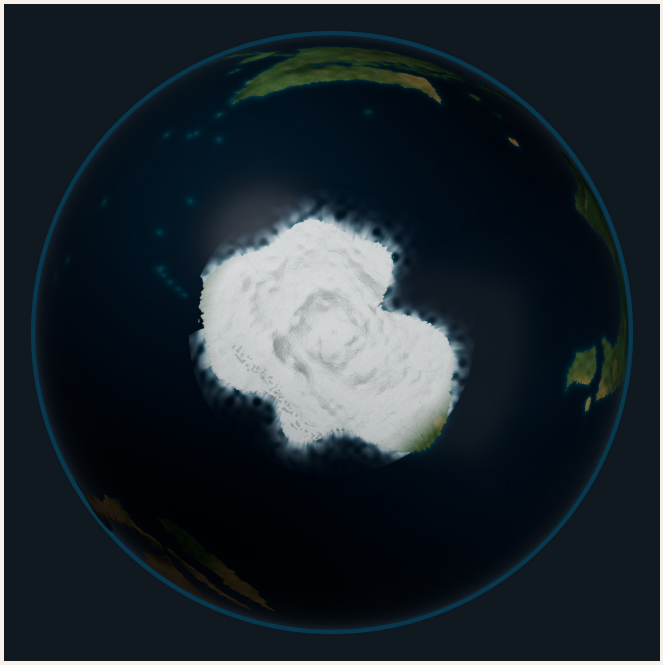
Connection to the deck

Earlier concepts propose an ancient route from Old Panterra to the newer cat country. This chain provides a candidate setting for that history while leaving the exact homeland, sea-level event and chronology open. [D39, D42]

Two poles with different identities



NORTH / OCEAN + ISLANDS



SOUTH / ICE CONTINENT

North: an ocean of routes and barriers

The Arctic region contains bedrock islands, glaciated uplands, channels and broken floating pack ice. The impact district lies on its continental margin. Leads and straits matter for navigation; ice does not form one featureless white latitude band.

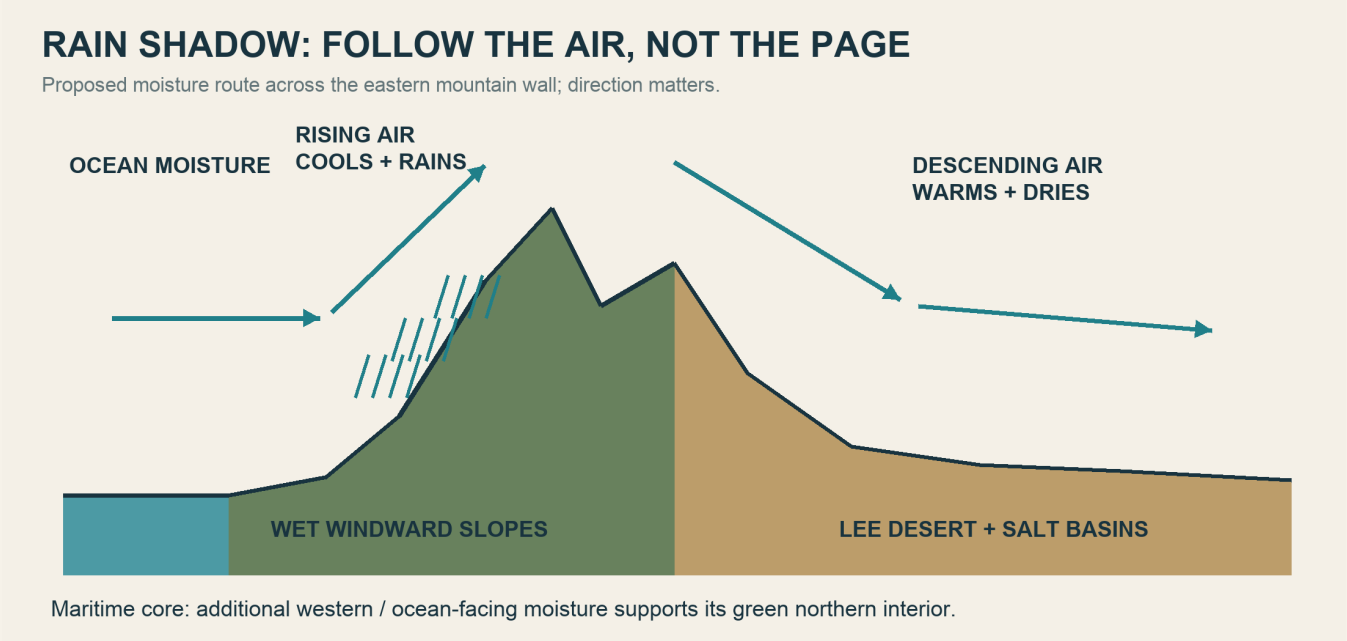
South: a separate ice continent

The southern pole lies within an asymmetric landmass with a peninsula, deep coastal embayments, mountain exposures and an interior ice dome. Grounded ice drains toward outlet glaciers and the surrounding shelves. It remains separate from the three inhabited continental cores.

What is fixed in the picture

The north-ocean / south-land contrast follows a useful Earth analogue, but the actual margins are proposals for this epoch. No ice-sheet or sea-ice simulation has been run. The 4,000-year orbit makes long-term advance and retreat important; this atlas is one moment in that cycle. [S5]

Climate follows terrain and exposure



Maritime core: additional western / ocean-facing moisture supports its green northern interior.

The mechanism

Air rising over a mountain barrier cools and can lose moisture as precipitation. Descending air on the lee side warms and dries. The relevant distinction is windward versus leeward, not left versus right on a flat drawing. [S2]

The circulation proposed here

Oblique storm tracks and monsoonal flow supply moisture from the northern/eastern marginal seas to the eastern wall. Its sheltered interior stays dry. Western and other ocean-facing exposure supplies the maritime core, keeping its northern interior greener. Panterra's western uplands intercept moisture and feed the Seattle catchment.

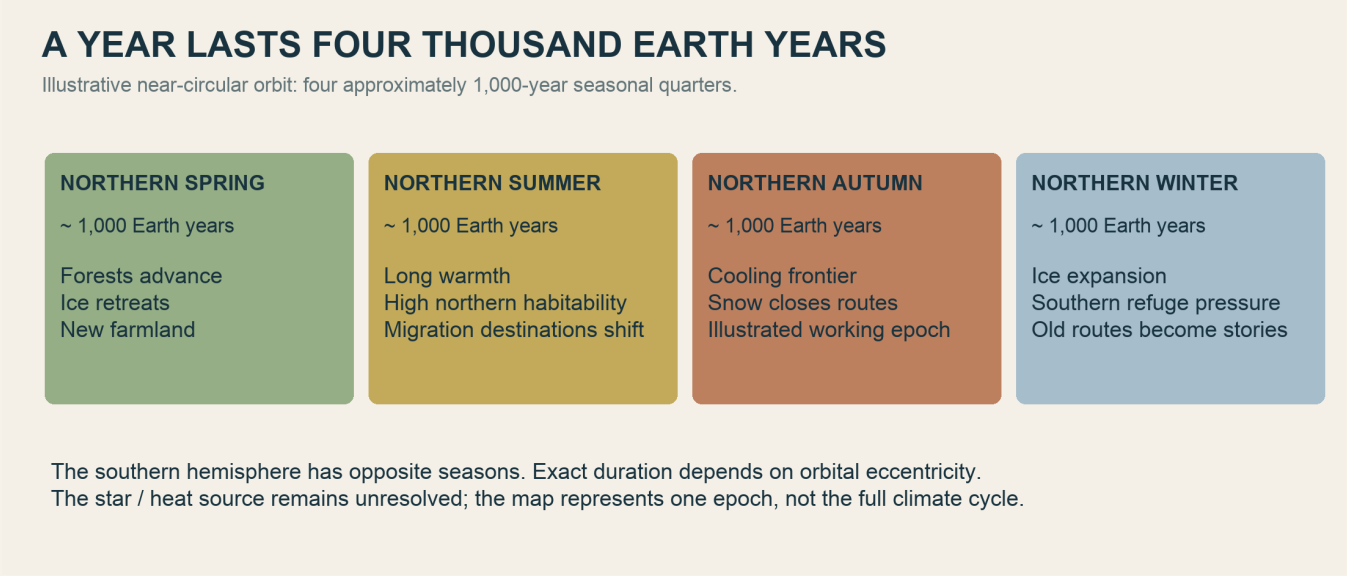
Transitions instead of painted zones

Forests give way to open woodland, grassland, rocky highlands and snow as moisture and elevation change. The maritime southern lee has drier pockets. Desert margins contain fans and seasonal channels. The biome field is an art-directed approximation, not the output of a global circulation model.

Do not freeze the whole cycle into one map

Long seasons can shift snowlines, storm paths, vegetation and coast access over generations. The displayed northern late-autumn epoch is a working assumption. Once the stellar premise is chosen, a climate specialist could test these proposed wind routes and revise them without discarding the geography.

Living inside thousand-year seasons



What 4,000 years means

With Earth-like tilt and a near-circular orbit, four seasonal quarters each last roughly 1,000 Earth years. Northern and southern seasons oppose each other. Eccentricity would make the quarters unequal. A 24-hour rotation is assumed for the model; the creator has not fixed it.

Consequences for history

A forest frontier can move while dynasties rise and fall. Farmland, ice margins and river regimes can change within inherited memory. Migration knowledge can become sacred history, and old routes may cease to be useful. Civilizations are not simply experiencing ordinary Earth weather in slow motion.

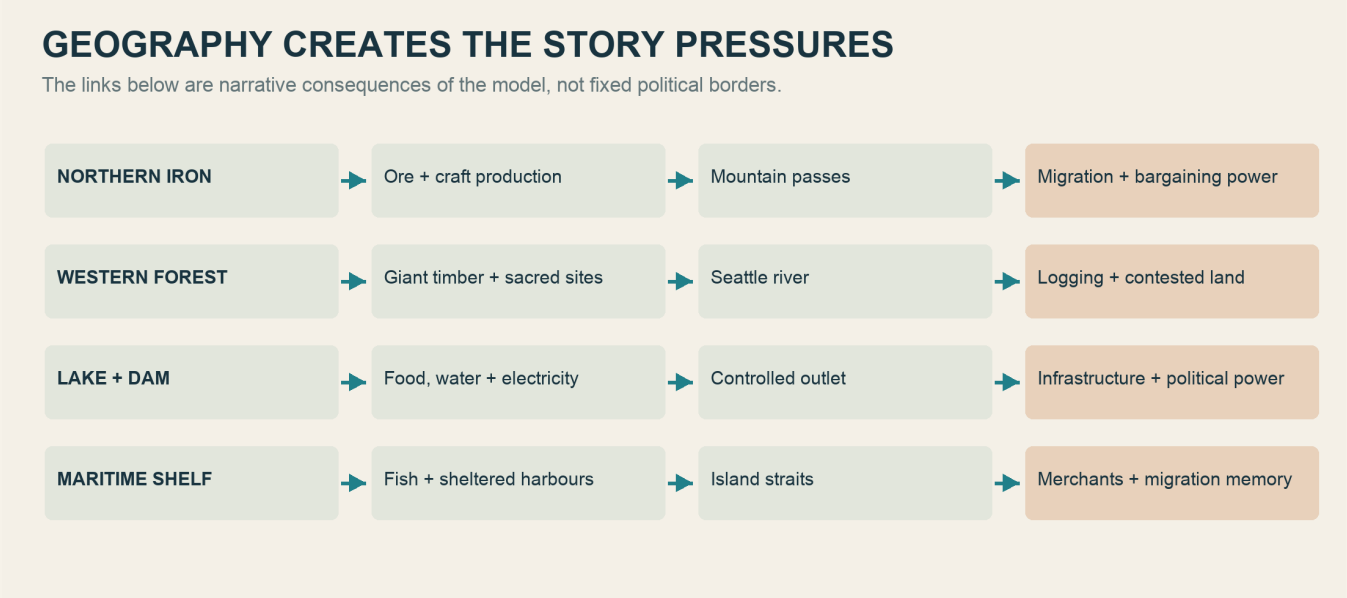
The heat-source problem remains open

Kepler's relation gives about 252 AU for a 4,000-year orbit around a one-solar-mass star. A Sun-like star would not supply Earth-like warmth there. The setting needs a different stellar/orbital or additional-heating premise. The atlas preserves the specified period rather than silently substituting an Earth year. [S6]

Industrial cooling in the draft

The deck discusses industrial activity accelerating northern cooling. Ordinary CO2 warming does not become an inverse greenhouse effect. Aerosols can produce cooling, but their composition, persistence and net balance must be specified. Treat causation as an unresolved story decision or an in-world disputed claim, not established climate physics. [S10; D4, D45]

Resources create routes and conflicts



The material contrast

The deck's Panterran economy relies on bronze/brass, scarce iron, waterways and electrical metallurgy. Northern birds bring access to an iron-rich province. The map supports that contrast through separate resource districts and costly passages between them. It does not require every useful metal to have the same geological origin. [D47]

The forest is contested infrastructure

Giant trees supply timber and support sacred avian sites. Rivers make those forests economically accessible to logging interests. Deforestation, damaged spawning streams and lost nesting places can therefore be local, visible causes of conflict rather than abstract background. [D4, D5]

The dam concentrates dependence

Power, lake level, downstream releases and transport all meet at one controlled outlet. Upstream fishing and farming communities, downstream merchants and industrial users depend on different aspects of its operation. The landscape supplies conflicting interests without prescribing who is right.

Limits of this atlas

Ore tonnages, forestry yields, crop output, electrical demand and transport capacity are not quantified. The map suggests where these systems interact; it is not a balanced economy simulation. Political ownership and final settlement names remain creator choices.

Cities and settlements need reasons

Seattle / upstream timber city

Place wharves, mills and storage beside accessible valley reaches, with forest roads above the floodplain. The city belongs to the primary river corridor. Its named global marker is a location anchor; buildings are not modeled on the globe.

London and Rome / lake country

London occupies southwestern lakeshore country and Rome a high eastern bank. Both can combine fisheries, lake shipping and agricultural hinterlands. Shorelines need harbours, terraces and seasonal operating levels before detailed city plans are drawn.

Amsterdam / downstream outlet

The large downstream city is a trade and processing center with canal links and alluvial land. Its wealth can come from controlling exchange and infrastructure rather than possessing rich local ore. This follows the deck's trade-outlet concept. [D41]

Milan / merchant island

The southern island is a useful site for maritime customs, resupply and finance. Its precise role in the inland supply chain still needs chosen routes. Do not assume the lake and sea share one uninterrupted navigable level.

Small places

Fishing villages belong in sheltered coves and on stable terraces. Crossing towns belong at ferries, bridges and junctions. Checkpoints belong at supply bottlenecks. Ancient ruins should predate the modern routes they overlook. The dam inset shows one village proposal; the planetary model does not contain a city-building kit.

Travel continuity

At 6,100 km radius, one degree of latitude is about 106 km. Several degrees between markers can mean weeks of travel, depending on transport and terrain. Route distances, seasonal pass closures and communication speeds must be set before simultaneous episode journeys are scripted.

Migration, sacred trees and choke points

A northern departure, not a featureless arrow

The bird homeland occupies the impact district and surrounding northern country in the draft. Cooling affects nesting sites, food supplies and accessible routes. Families, livestock, freight and institutions do not move like a single airborne flock. [D2, D5]

Why the passes matter to birds

Flying individuals can cross barriers that stop carts, but high altitude, weather, loads, feeding stops and ground-based supply still create preferred corridors. Western, Central and Eastern Gate are proposed lower passages. They are logistical choke points, not an absolute biological wall.

Where control becomes visible

Checkpoint stations can cluster at valley mouths, crossings, staging fields and supply depots. Beyond them, routes branch toward timber country, lakeshore towns and the downstream metropolis. The model locates candidate gates; it does not specify a border regime or a completed road network.

The forest preserves memory

Ancient avian carvings and cliff ruins sit above the western forest corridor. The old-growth refuge provides a setting where material evidence, oral history and present land claims meet. Logging can expose ruins while destroying the living landscape around them. Those are narrative proposals grounded in the deck's sacred-tree material. [D5, D9]

Chronology is deliberately unfinished

The deck mentions roughly 700 years of northern occupation, millennial migrations and possible early departure. A 4,000-year orbit does not automatically imply a migration every 1,000 years. Decide which seasonal threshold drives movement, whether migrations alternate hemispheres, and which dates are history versus tradition. [D2, D5, D45]

Use the globe as a drawing reference

Choose the angle before drawing

Open the offline atlas and select the hemisphere or regional view. Orthographic cameras show the actual sphere without a map-projection stretch. Land near the limb is foreshortened; choose a closer-centered view when a region is the drawing subject. The annotated versions locate features on the same renders.

Open the editable model

CATWORLD_Terra.blend contains the terrain, packed textures, named landmark anchors and twelve atlas cameras. Frames 1-12 select their cameras and matching lights. The separate dam file uses kilometre units and two local views. Do not merge the files without converting their different scales.

Resolution and exaggeration

The global surface uses 4096 x 2048 textures: roughly 9.4 km per texel at the equator. The mesh has about 1.18 million vertices. Relief is displayed at 10x; the primary river is deliberately widened. Houses, individual trees, narrow channels and the dam wall need local models.

What can be edited

Feature names and coordinates are in features.json. Primary flow paths are in hydrology_paths.json. The base sketch reconstruction and R4 changes are in build_terrain.py and reshape_geography.py. Camera definitions are in build_blender.py. The source scripts allow reproducible revision; the packed blend is usable without rebuilding.

Light and atmosphere

Clouds are omitted to expose the terrain. Each camera has broad daylight from a useful drawing angle, not one simultaneous astronomical illumination state. The blue limb is a restrained visual atmosphere; it is not an atmospheric scattering simulation.

Decisions for the creator to make next

1 / Stellar premise and seasonal epoch

Keep the 4,000-year orbit, then decide how the planet receives habitable heat. Confirm whether the story starts in northern late autumn. This controls the long-term climate and migration logic.

2 / Canon geography and names

Approve or revise the three-core arrangement, the new maritime interior, the exact location of Old Panterra and the former land bridge. Replace Seattle, London, Rome, Amsterdam and Milan shorthand with final names when ready.

3 / Reservoir history

Choose the pre-dam lake extent, dam construction sequence, filling timescale, operating level and downstream obligations. Decide whether the super-lake's extreme scale is literal or should be reduced before local maps are commissioned.

4 / Energy and material limits

Choose reliable rather than nominal power output, catchment assumptions, industrial demand and the role of scarce iron. Treat naturally enriched uranium and any unusual cooling mechanism as separate fictional premises requiring decisions.

5 / Routes and chronology

Fix pass accessibility, boat/canal connections, shipping speeds and journey distances. Reconcile the deck's migration intervals with the seasonal cycle. Distinguish old records, oral tradition and political claims.

6 / Next level of production detail

After those choices, commission local maps of the Seattle corridor, lake shores, dam district and selected gate valleys. They should inherit this globe's direction and scale, while replacing its exaggerated river widths and simplified settlement markers.

Sources and reference notes

D / Project references

CATWORLD PITCH DECK.pdf, 50-page development draft, and map sketches 3-5. D2 = overview/migration; D4 = Panterra; D5 = Flock; D9 = Cyrus context; D12 = Pieter/dam village; D38-42 = map concepts; D45 = chronology; D47 = materials and dam. Page numbers refer to PDF pages. The latest creator corrections take precedence. Private message screenshots and the deck itself are not redistributed in this package.

S1 / Collision mountains

USGS, The Himalayas, https://pubs.usgs.gov/gip/dynamic/himalaya.html ; National Park Service, Collisional Mountain Ranges, https://www.nps.gov/subjects/geology/plate-tectonics-collisional-mountain-ranges.htm

S2 / Rain shadows

NOAA, Teek and Tom educator guide, lesson 2, https://oceantoday.noaa.gov/teekandtom/educators-guide/lesson-2.pdf

S3 / Red iron oxides

USGS, Find-A-Feature: Red Rocks, https://www.usgs.gov/educational-resources/find-feature-red-rocks

S4 / Impact-related resources

Reimold and colleagues, Economic mineral deposits in impact structures: A review (2005), abstract record at https://hero.epa.gov/reference/1435484/ . The resource mechanisms are analogues, not a measured ore model for Terra.

S5 / Polar analogues

National Snow and Ice Data Center: https://nsidc.org/learn/parts-cryosphere/sea-ice and https://nsidc.org/learn/parts-cryosphere/ice-sheets

Sources and reference notes / continued

S6 / Orbital period

NASA, Orbits and Kepler's Laws, https://science.nasa.gov/solar-system/orbits-and-keplers-laws/ . The 252 AU figure is this guide's calculation for a one-solar-mass primary.

S7 / Hydropower

US Department of Energy, How Hydropower Works, https://www.energy.gov/cmei/water/how-hydropower-works . All Terra discharge, head, efficiency and power values are explicitly assumed examples.

S8-S9 / Dam scale comparisons

US Bureau of Reclamation, Hoover Dam, https://www.usbr.gov/projects/index.php?id=122 ; China Energy Engineering Corporation project profile, Three Gorges maximum height and capacity, https://en.ceec.net.cn/attach/0/1903261138048558920.pdf

S10 / Warming and aerosols

NASA, Causes of Climate Change, https://science.nasa.gov/climate-change/causes/ ; NASA, Aerosols: Small Particles with Big Climate Effects, https://science.nasa.gov/science-research/earth-science/climate-science/aerosols-small-particles-with-big-climate-effects/

T / Terrain provenance

Mapzen Terrain Tiles on AWS, with USGS SRTM/GMTED elevation sources. A Himalayan mosaic was warped, rescaled and combined with fictional terrain. https://registry.opendata.aws/terrain-tiles/ . Full provider attribution is bundled in source/TERRAIN_ATTRIBUTION.md. No provider endorses this fictional model.

Landmark catalogue / 1 of 3

01 / Ferric Sea / impact basin

64.987 latitude, 1.817 longitude. Ancient impact; iron-rich sediment and central uplift

02 / Northern Divide

36.819 latitude, -6.658 longitude. Enormous folded belt; multiple spines and high plateaus

03 / Panterran Plateau

21.091 latitude, 3.863 longitude. Dissected upland below the mountain wall

04 / Great Reservoir

11.504 latitude, -10.830 longitude. Flooded continental basin, maintained by the dam

05 / Great Panterran Dam

5.133 latitude, -2.476 longitude. Reservoir outlet at a bedrock gorge

06 / Alluvial trade outlet

-4.402 latitude, 10.242 longitude. Amsterdam shorthand; delta, canals and commerce

07 / Old-growth refuge

21.091 latitude, -38.013 longitude. Wet uplands, giant trees and logging frontier

08 / Western Gate

30.249 latitude, -46.788 longitude. Lower mountain saddle; migration checkpoint

09 / Central Gate

35.517 latitude, 22.508 longitude. Second low passage through Northern Divide

10 / Eastern Gate

38.115 latitude, 64.587 longitude. Eastern coastal and ridge crossing

Landmark catalogue / 2 of 3

11 / Maritime river country

38.000 latitude, -158.000 longitude. Forested northern foreland with branching rivers and coastal plains

12 / Maritime great range

23.000 latitude, -164.000 longitude. Third continental core joins the global collision belt

13 / Drowned land bridge

-12.000 latitude, -57.000 longitude. Island chain and former low-sea-level migration route

14 / Pieter’s fishing village

5.147 latitude, -2.443 longitude. Local dam inset site; terrace location is below global texture resolution

15 / Ancient avian cliff ruins

23.627 latitude, -47.014 longitude. Ruins above the western forest corridor

16 / London / lakeshore country

8.306 latitude, -13.931 longitude. Proposed city on the southwestern lake shore; elevated above the reservoir

17 / Merchant island

-31.190 latitude, 7.219 longitude. Milan shorthand; trade controls in southern strait

18 / Great leeward desert

-0.314 latitude, 120.722 longitude. Dry interior on the left of the marked far-side view

19 / Eastern rift gulf

11.737 latitude, 92.865 longitude. Fault-bounded gulf opening into the southern sea

20 / Eastern marginal gulf

39.000 latitude, -131.000 longitude. Drowned failed-rift basin and broken capes

Landmark catalogue / 3 of 3

21 / Old maritime shield

41.000 latitude, -148.000 longitude. Old shield with deeply weathered forests and river plains

22 / Eastern collision knot

32.896 latitude, 70.405 longitude. Broad transpressional uplands joining eastern terranes

23 / Northern fjord remnant

70.860 latitude, -172.792 longitude. Glacial valleys cut into the high-latitude remnant

24 / Seattle / upstream timber city

20.409 latitude, -33.549 longitude. Primary river flows from here INTO the super-lake

25 / Rome / eastern lake shore

10.679 latitude, -2.593 longitude. Proposed high-bank city overlooking the lake; distinct from the downstream metropolis

26 / Compact collision junction

16.000 latitude, 167.000 longitude. Oblique junction where three continental blocks meet through broad ranges

27 / Drowned shelf archipelago

-24.000 latitude, -87.000 longitude. Former land bridge toward Panterra; submerged saddle chain

28 / Arctic crown islands

80.000 latitude, -80.000 longitude. Glaciated bedrock islands and straits amid broken pack ice

29 / South polar ice continent

-77.000 latitude, 105.000 longitude. Asymmetric land ice, coastal mountains and outlet glaciers

30 / Southern embayment

-76.000 latitude, -25.000 longitude. Deep marine inlet interrupts the south polar ice sheet

Specifications and camera index

Model scale

Radius 6,100 km; axial tilt 23.4 degrees; orbital period 4,000 Earth years; assumed rotation 24 hours. Globe radius in Blender is 10 units. Physical elevation is separate from 10x display relief. The dam model uses 1 unit = 1 km.

Resolution

1,181,953 globe vertices and 1,179,648 quads. Four packed 4096 x 2048 maps. Twelve 2048 x 2048 globe renders and two 2400 x 1600 dam renders. Thirty named geographic anchors. Reservoir footprint: 1,535,016 km2. Peak physical elevation: 10.74 km.

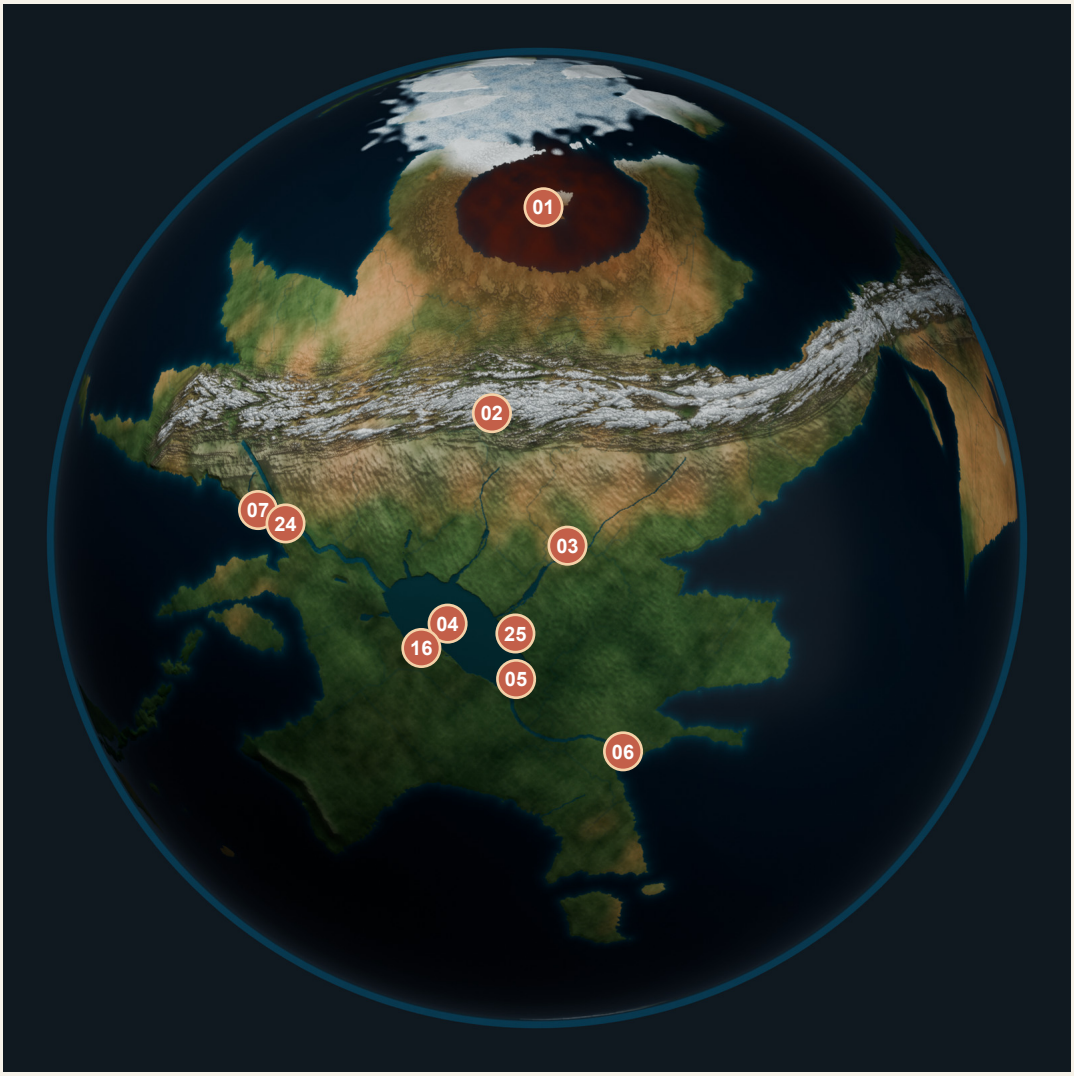
Validation

Three main cores are connected; the south-polar land is separate. Sampled primary river and outlet grades descend. Named city sites are checked against land and reservoir masks. Source scripts, image sizes, gallery links and packed textures are checked. These checks do not replace tectonic, climate, erosion or hydraulic simulation.

Frame 01 - 01_panterra - center 22 lat / 0 lon
Frame 02 - 02_eastern - center 0 lat / 90 lon
Frame 03 - 03_western - center 0 lat / -90 lon
Frame 04 - 04_far_side - center 0 lat / 180 lon
Frame 05 - 05_north_pole - center 90 lat / 0 lon
Frame 06 - 06_south_pole - center -90 lat / 0 lon
Frame 07 - 07_panterra_close - center 15 lat / -8 lon
Frame 08 - 08_impact_close - center 65 lat / 2 lon
Frame 09 - 11_maritime_continent - center 35 lat / -153 lon
Frame 10 - 12_eastern_continent - center 5 lat / 135 lon
Frame 11 - 13_seattle_waterway - center 16 lat / -25 lon
Frame 12 - 14_archipelago - center -12 lat / -87 lon

Dam file: frame 1 = valley context; frame 2 = structural detail. Clean PNGs, annotated PNGs, editable models, textures, source scripts, CSV landmarks and this guide are included in the full archive.

Panterra / front hemisphere



Center 22 latitude / 0 longitude. Orthographic view; relief displayed at 10x.

01 / Ferric Sea / impact basin
Ancient impact; iron-rich sediment and central uplift

02 / Northern Divide
Enormous folded belt; multiple spines and high plateaus

03 / Panterran Plateau
Dissected upland below the mountain wall

04 / Great Reservoir
Flooded continental basin, maintained by the dam

05 / Great Panterran Dam
Reservoir outlet at a bedrock gorge

06 / Alluvial trade outlet
Amsterdam shorthand; delta, canals and commerce

07 / Old-growth refuge
Wet uplands, giant trees and logging frontier

16 / London / lakeshore country
Proposed city on the southwestern lake shore; elevated above the reservoir

24 / Seattle / upstream timber city
Primary river flows from here INTO the super-lake

25 / Rome / eastern lake shore
Proposed high-bank city overlooking the lake; distinct from the downstream metropolis

Eastern hemisphere



Center 0 latitude / 90 longitude. Orthographic view; relief displayed at 10x.

10 / Eastern Gate
Eastern coastal and ridge crossing

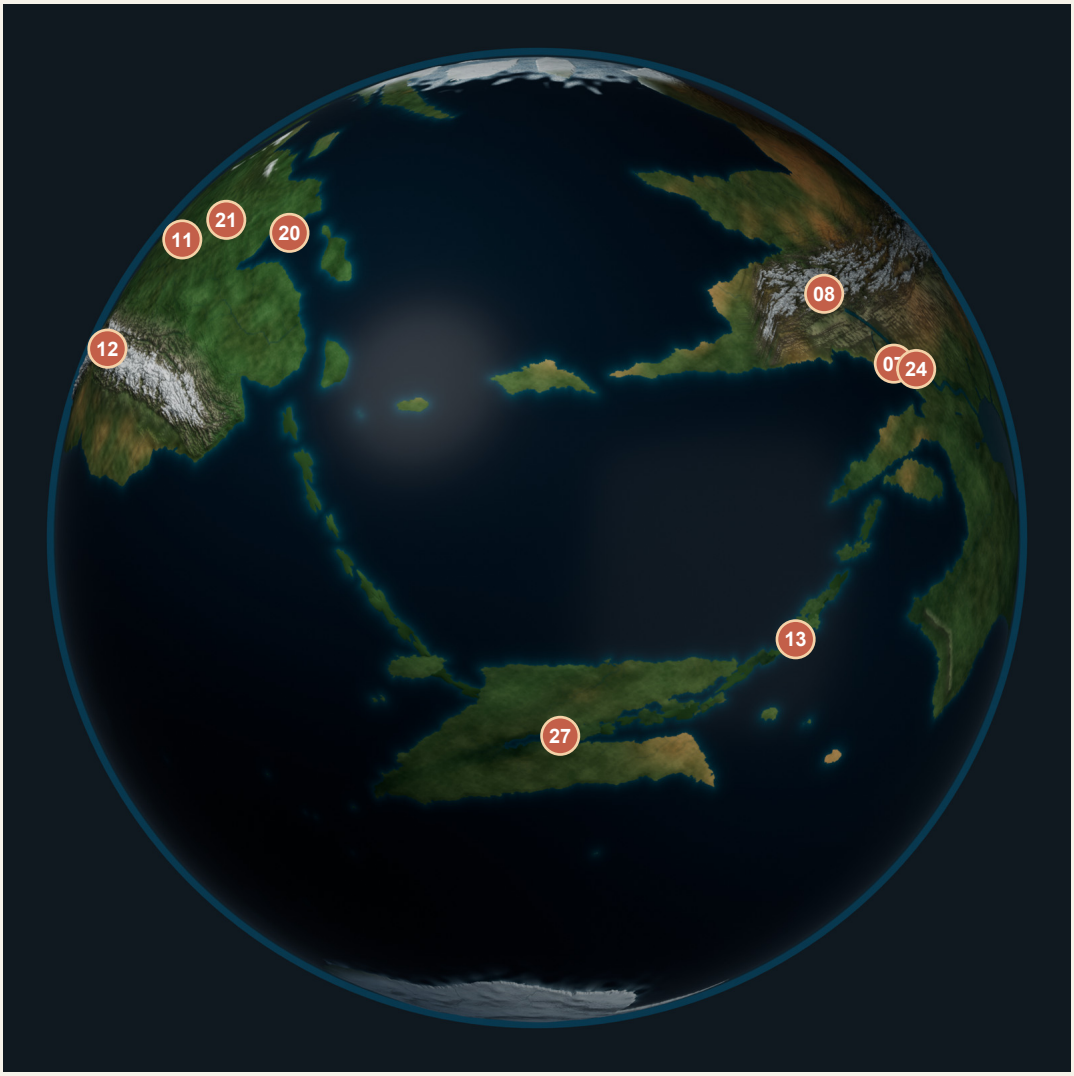
19 / Eastern rift gulf
Fault-bounded gulf opening into the southern sea

26 / Compact collision junction
Oblique junction where three continental blocks meet through broad ranges

18 / Great leeward desert
Dry interior on the left of the marked far-side view

22 / Eastern collision knot
Broad transpressional uplands joining eastern terranes

Western hemisphere



Center 0 latitude / -90 longitude. Orthographic view; relief displayed at 10x.

07 / Old-growth refuge
Wet uplands, giant trees and logging frontier

08 / Western Gate
Lower mountain saddle; migration checkpoint

11 / Maritime river country
Forested northern foreland with branching rivers and coastal plains

12 / Maritime great range
Third continental core joins the global collision belt

13 / Drowned land bridge
Island chain and former low-sea-level migration route

20 / Eastern marginal gulf
Drowned failed-rift basin and broken capes

21 / Old maritime shield
Old shield with deeply weathered forests and river plains

24 / Seattle / upstream timber city
Primary river flows from here INTO the super-lake

27 / Drowned shelf archipelago
Former land bridge toward Panterra; submerged saddle chain

Far-side hemisphere



Center 0 latitude / 180 longitude. Orthographic view; relief displayed at 10x.

11 / Maritime river country
Forested northern foreland with branching rivers and coastal plains

12 / Maritime great range
Third continental core joins the global collision belt

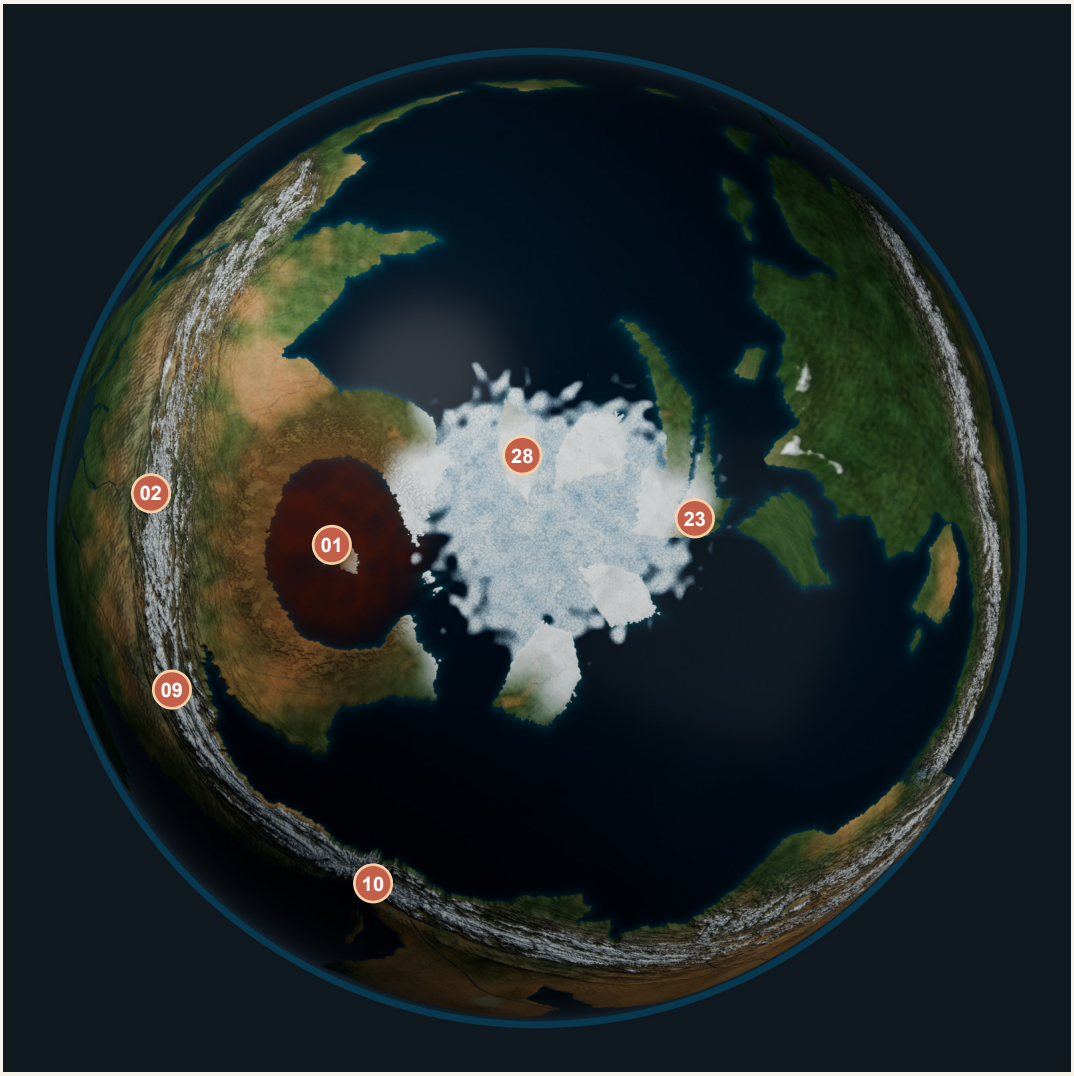
18 / Great leeward desert
Dry interior on the left of the marked far-side view

21 / Old maritime shield
Old shield with deeply weathered forests and river plains

23 / Northern fjord remnant
Glacial valleys cut into the high-latitude remnant

26 / Compact collision junction
Oblique junction where three continental blocks meet through broad ranges

North polar view



Center 90 latitude / 0 longitude. Orthographic view; relief displayed at 10x.

01 / Ferric Sea / impact basin
Ancient impact; iron-rich sediment and central uplift

02 / Northern Divide
Enormous folded belt; multiple spines and high plateaus

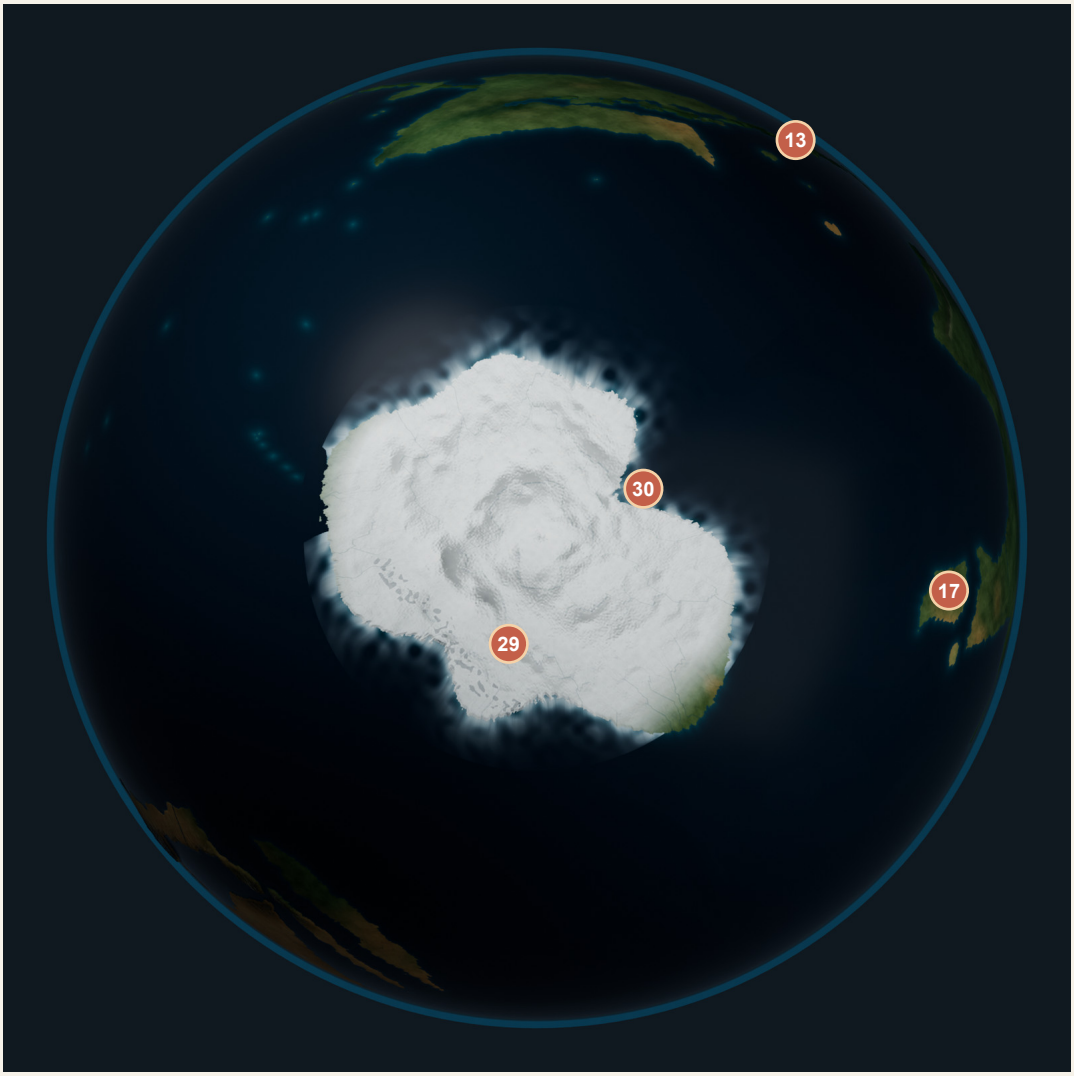
09 / Central Gate
Second low passage through Northern Divide

10 / Eastern Gate
Eastern coastal and ridge crossing

23 / Northern fjord remnant
Glacial valleys cut into the high-latitude remnant

28 / Arctic crown islands
Glaciated bedrock islands and straits amid broken pack ice

South polar view



Center -90 latitude / 0 longitude. Orthographic view; relief displayed at 10x.

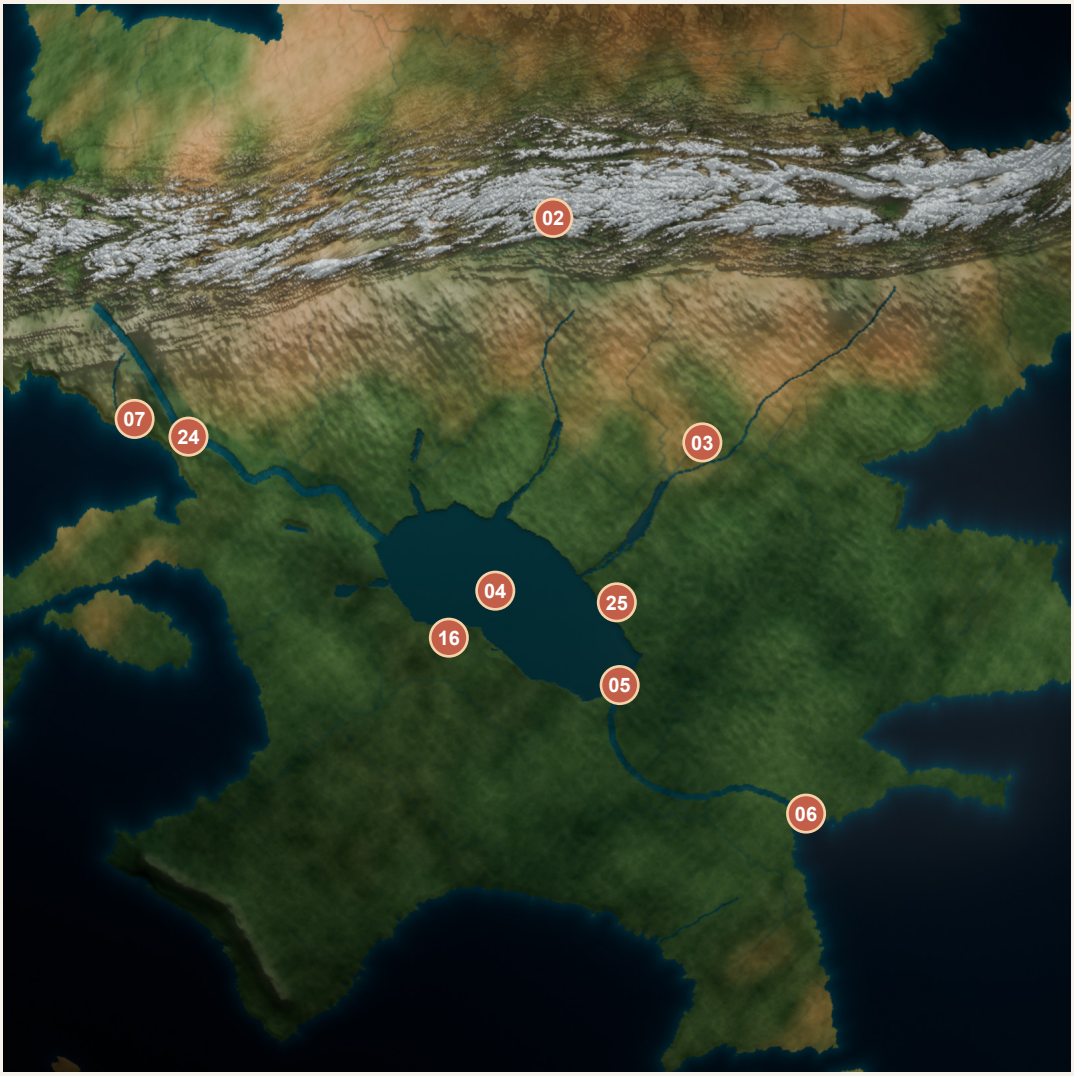
13 / Drowned land bridge
Island chain and former low-sea-level migration route

17 / Merchant island
Milan shorthand; trade controls in southern strait

29 / South polar ice continent
Asymmetric land ice, coastal mountains and outlet glaciers

30 / Southern embayment
Deep marine inlet interrupts the south polar ice sheet

Panterra / regional study



Center 15 latitude / -8 longitude. Orthographic view; relief displayed at 10x.

FLOW: Seattle -> super-lake -> dam -> downstream city

02 / Northern Divide
Enormous folded belt; multiple spines and high plateaus

03 / Panterran Plateau
Dissected upland below the mountain wall

04 / Great Reservoir
Flooded continental basin, maintained by the dam

05 / Great Panterran Dam
Reservoir outlet at a bedrock gorge

06 / Alluvial trade outlet
Amsterdam shorthand; delta, canals and commerce

07 / Old-growth refuge
Wet uplands, giant trees and logging frontier

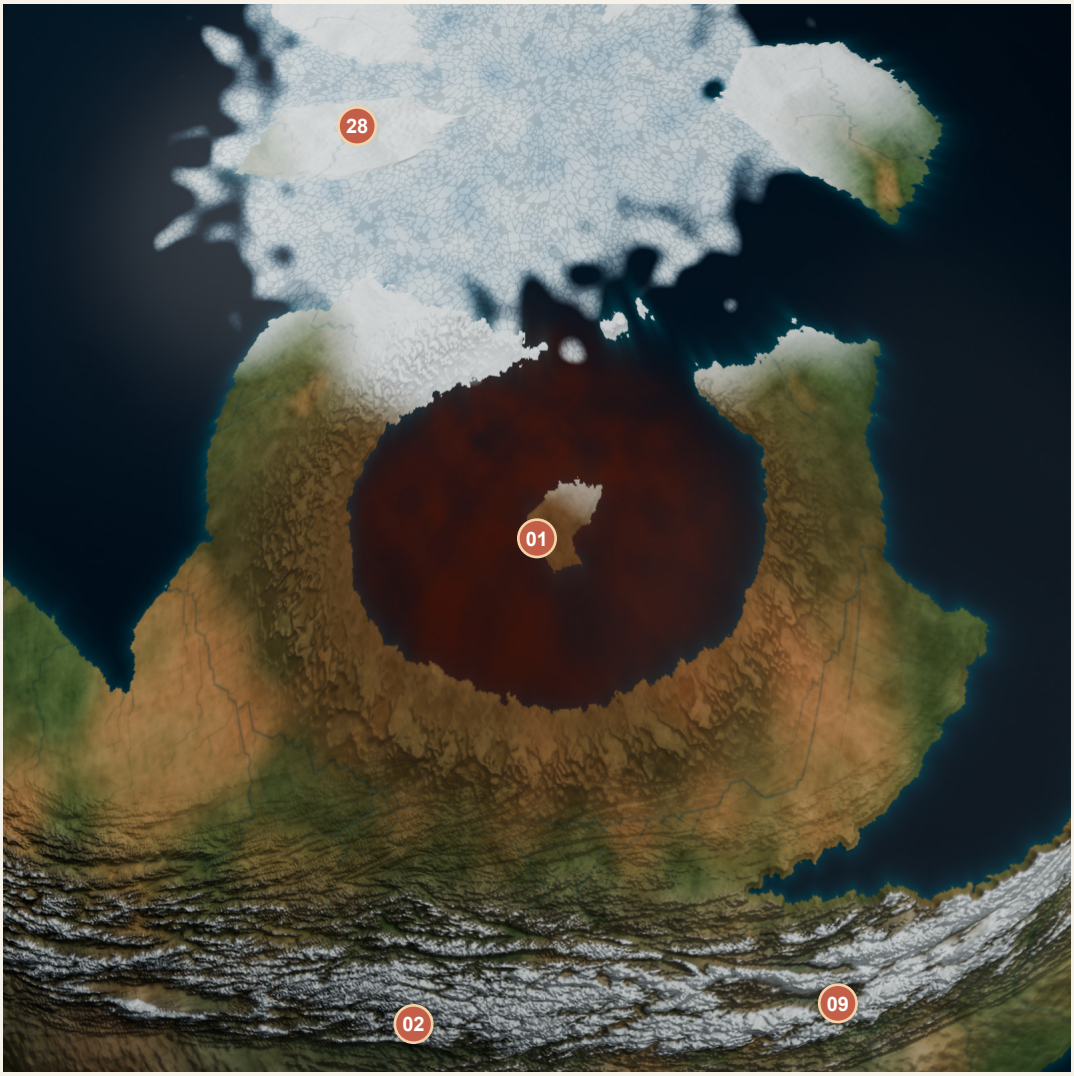
16 / London / lakeshore country
Proposed city on the southwestern lake shore; elevated above the reservoir

24 / Seattle / upstream timber city
Primary river flows from here INTO the super-lake

25 / Rome / eastern lake shore
Proposed high-bank city overlooking the lake; distinct from the downstream metropolis

REFERENCE PLATE 08

Northern impact basin



Center 65 latitude / 2 longitude. Orthographic view; relief displayed at 10x.

01 / Ferric Sea / impact basin
Ancient impact; iron-rich sediment and central uplift

02 / Northern Divide
Enormous folded belt; multiple spines and high plateaus

09 / Central Gate
Second low passage through Northern Divide

28 / Arctic crown islands
Glaciated bedrock islands and straits amid broken pack ice

Great dam / valley context



Local engineering concept. Scale: 1 unit = 1 km.

A / Great Reservoir

B / Gravity dam / service crest

C / Gated spillway

D / Turbine galleries

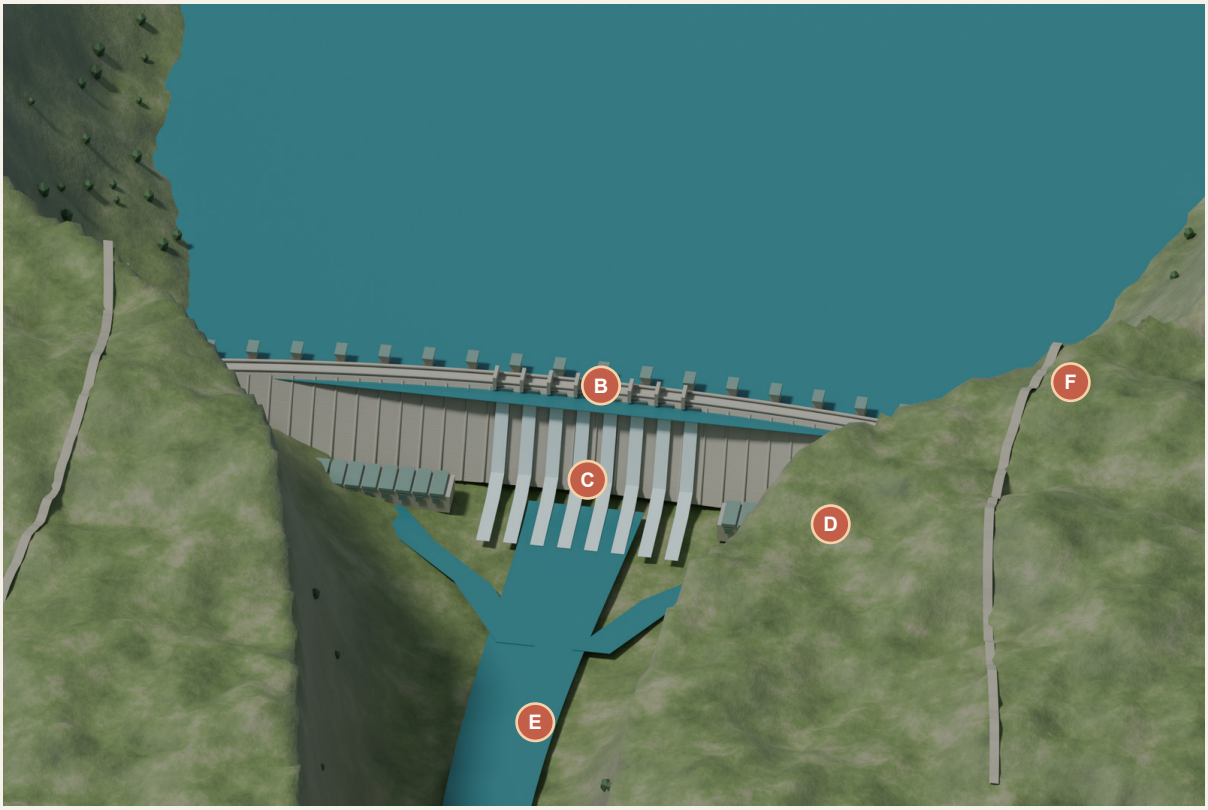
E / Tailwater / river

F / Abutment checkpoint

G / Fishing village

REFERENCE PLATE 10

Great dam / structural study



Local engineering concept. Scale: 1 unit = 1 km.

B / Gravity dam / service crest

C / Gated spillway

D / Turbine galleries

E / Tailwater / river

F / Abutment checkpoint

Maritime continent



Center 35 latitude / -153 longitude. Orthographic view; relief displayed at 10x.

11 / Maritime river country
Forested northern foreland with branching rivers and coastal plains

12 / Maritime great range
Third continental core joins the global collision belt

20 / Eastern marginal gulf
Drowned failed-rift basin and broken capes

21 / Old maritime shield
Old shield with deeply weathered forests and river plains

26 / Compact collision junction
Oblique junction where three continental blocks meet through broad ranges

Eastern continent



Center 5 latitude / 135 longitude. Orthographic view; relief displayed at 10x.

10 / Eastern Gate
Eastern coastal and ridge crossing

19 / Eastern rift gulf
Fault-bounded gulf opening into the southern sea

18 / Great leeward desert
Dry interior on the left of the marked far-side view

26 / Compact collision junction
Oblique junction where three continental blocks meet through broad ranges

Seattle / primary waterway



Center 16 latitude / -25 longitude. Orthographic view; relief displayed at 10x.

FLOW: Seattle -> super-lake -> dam -> downstream city

04 / Great Reservoir
Flooded continental basin, maintained by the dam

16 / London / lakeshore country
Proposed city on the southwestern lake shore; elevated above the reservoir

07 / Old-growth refuge
Wet uplands, giant trees and logging frontier

24 / Seattle / upstream timber city
Primary river flows from here INTO the super-lake

Drowned land bridge



Center -12 latitude / -87 longitude. Orthographic view; relief displayed at 10x.

11 / Maritime river country
Forested northern foreland with branching rivers and coastal plains

20 / Eastern marginal gulf
Drowned failed-rift basin and broken capes

13 / Drowned land bridge
Island chain and former low-sea-level migration route

27 / Drowned shelf archipelago
Former land bridge toward Panterra; submerged saddle chain